Image
Text
Audio
Feature extractor
Feature extractor
Feature extractor
Fusion operator set: F
Feature set: M
MBTF search space
Encode
I_1
I_2
I_{k-1}
I_k
Fitness assessment
Population initialization
Population fitness
Selection
End?
Yes
Best individual
No
Offspring generation
Initial population
Zero-cost evaluation
Top-ρ
Candidate population
Remaining
Unqualified population
Fitness assessment
Fitness=0
Fitness=0.9525
Fitness=0.9527
Fitness=0
Fitness=0
Evaluated population
The screening-then-training strategy
Screening
Training
Zero-cost scoring for candidate architectures
Sampled individual
Decode
Forward pass
Backbone modules
Progressivity analysis
Feature centering
Covariance matrix
Expressivity score (Overall feature diversity)
Progressivity score (Depth-wise diversity growth)
Zero-cost architecture scores